

Customer Conversion Prediction Using Machine Learning

Objective

The objective of this project was to develop a machine learning model capable of predicting whether a customer would convert based on demographic information, browsing behavior, and historical interaction data. The project followed a complete machine learning workflow including data exploration, preprocessing, model training, evaluation, and submission generation.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the structure and quality of the dataset. Various features such as City Tier, Device Type, Traffic Source, Pages Viewed, Products Viewed, Previous Purchases, Browser Version, and Time on Site were analyzed to identify their relationship with customer conversion behavior.

Missing value analysis revealed incomplete observations in numerical features including Age and Income. Distribution analysis was performed using histograms and skewness measurements. Both Age and Income exhibited near-symmetric distributions, indicating that mean imputation would be an appropriate strategy for handling missing values.

Data Preprocessing

Several preprocessing steps were performed to prepare the dataset for model training:

- The target variable was identified as **Converted**.
- Categorical variables were encoded into numerical representations to enable model compatibility.
- Missing values in numerical features were handled using **SimpleImputer** with mean imputation.
- Relevant numerical features were selected for training.
- The dataset was split into training and validation sets using an 80:20 ratio with stratified sampling to preserve class distribution.
- Feature scaling was performed using **StandardScaler** to normalize feature values and improve model convergence.

Model Development

A **Logistic Regression** classifier was selected as the primary model due to its effectiveness, interpretability, and suitability for binary classification problems.

Model configuration:

- Algorithm: Logistic Regression
- Maximum Iterations: 1000
- Random State: 42

The model was trained using the preprocessed and scaled training dataset.

Results

Model performance was evaluated using both training and validation datasets.

- Training Accuracy: **72.25%**
- Validation Accuracy: **73.33%**

The validation accuracy exceeded the training accuracy slightly, indicating that the model generalized well to unseen data and did not exhibit signs of significant overfitting. The model demonstrated a reliable ability to identify customer conversion patterns based on the available features.

After evaluation, the trained model was used to generate predictions for the private test dataset. The predictions were formatted according to the competition requirements and exported as a submission CSV file.

Conclusion

This project successfully implemented an end-to-end machine learning pipeline for customer conversion prediction. Through exploratory data analysis, preprocessing, feature transformation, and Logistic Regression modeling, a robust predictive system was developed and evaluated. The final model achieved a validation accuracy of 73.33% and was used to generate competition-ready predictions for submission.